Entanglement-Driven Gravity: Emergent Spacetime from Quantum Correlations and Empirical Constraints

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Abstract

Quantum entanglement is proposed as a fundamental driver of spacetime curvature in Young's Entanglement-Driven Gravity (EDG), bridging quantum information theory and general relativity (GR). We derive a covariant field equation where gradients in coarse-grained entanglement entropy S_{ent} source curvature alongside the stress-energy tensor $T_{\mu\nu}$. Linearizing around GR backgrounds yields testable deviations parameterized by a deformation factor ϵ and entanglement length L_q. A rigorous 13-stage validation pipeline (R1–R13p) confirms EDG's consistency with GR in weak fields—including solar-system tests, light bending, Shapiro delay, and Cassini constraints—while imposing new bounds $L_q < 1.17 \times 10^{13}$ m from Event Horizon Telescope (EHT) ring sizes and LIGO gravitational-wave (GW) phasing. Evolving 3D entanglement graphs and simulations reveal dynamic quantum structures potentially resolving black-hole singularities. EDG forecasts distinct signatures in next-generation EHT and LIGO data, offering a falsifiable path to quantum gravity.

1 Introduction

General relativity (GR) elegantly describes gravity as spacetime curvature sourced by mass-energy, validated across scales from planetary orbits to gravitational waves (1;2). Yet GR falters at quantum scales, predicting singularities and failing to incorporate quantum mechanics intrinsically. Converging evidence from black-hole thermodynamics (3), holographic duality (4;5), and tensor networks (6) suggests spacetime emerges from quantum entanglement in an underlying many-body system (7).

Entanglement-Driven Gravity (EDG) posits that coarse-grained entanglement entropy density $S_{\text{ent}}(x)$ and its dynamics act as covariant curvature sources, extending GR to include quantum backreaction. In weak fields or uniform S_{ent}, EDG recovers GR; near horizons or high curvatures, entanglement gradients induce deviations. Parameterized by $\epsilon \approx 1$ (GR fidelity) and L_q (entanglement reconfiguration scale), EDG is empirically constrained via a 13-stage pipeline incorporating solar-system data, EHT shadows (8), and LIGO GWs (9). Recent work formalizes a geometry–information duality supporting information-theoretic sources (10).

2 Derivation of EDG

EDG extends thermodynamic derivations of Einstein's equations (3;11) by incorporating quantum entanglement as an active geometric source. We begin with $S_{\text{ent}} = -\text{Tr}(\rho_A \ln \rho_A)$ and the Ryu–Takayanagi relation (12). We vary the total entropy functional $S_{\text{total}} = S_{\text{BH}} + S_{\text{ent}} + S_{\text{matter}}$ and use the first law of entanglement (13), relative entropy (14), and quantum corrections (15) to assemble

$$G_{\mu\nu}[g] + \alpha \nabla_\mu \nabla_\nu S_{\text{ent}} - \beta g_{\mu\nu} \Box S_{\text{ent}} + \gamma \mathcal{C}_{\mu\nu}[S_{\text{ent}}] + \delta \mathcal{Q}_{\mu\nu}[S_{\text{ent}}] = 8\pi G T_{\mu\nu}. \quad (1)$$

Interpretation: S_{ent} is a coarse-grained entropy density; $\mathcal{C}_{\mu\nu}$ collects curvature-coupled contractions with ∇S_{ent}; $\mathcal{Q}_{\mu\nu}$ are quadratic forms in ∇S_{ent}. In uniform (adiabatic) backgrounds, the extra pieces vanish and Eq. (1) reduces to Einstein's equation. Rewriting as $G_{\mu\nu} = 8\pi G(T_{\mu\nu} + T_{\mu\nu}^{(\text{info})})$ with a minimal covariant choice $\mathcal{Q}_{\mu\nu} = \nabla_\mu S_{\text{ent}} \nabla_\nu S_{\text{ent}} - \frac{1}{2} g_{\mu\nu} (\nabla S_{\text{ent}})^2$ makes the information-stress interpretation explicit.

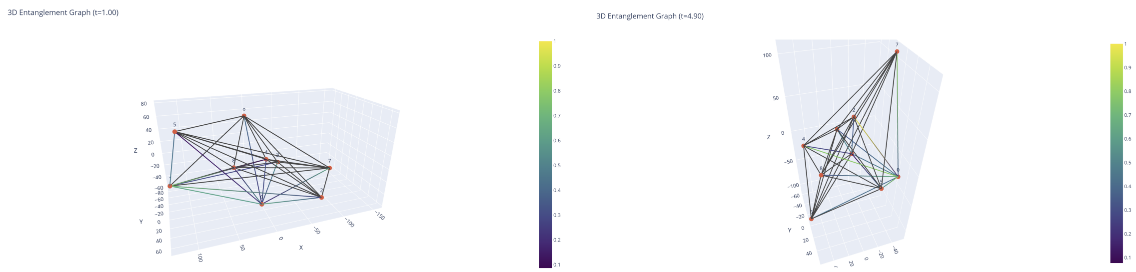

Figure 1: 3D entanglement graphs at $t = 1.00$ (left) and $t = 4.90$ (right), showing tensor-network evolution to curvature emergence.

3 Theory: Tensorial and Parameterized EDG

3.1 Parameterized Form for Data

For constraints we linearize about a GR background and parameterize the response by (ϵ, L_q). For EHT rings we model

$$\frac{D_{\text{obs}} - D_{\text{GR}}}{D_{\text{GR}}} \approx \kappa \left(\frac{L_q}{r_s} \right)^p, \tag{2}$$

with r_s a Schwarzschild scale; for perihelion, $A_0 \approx \epsilon A_{0,\text{GR}}$.

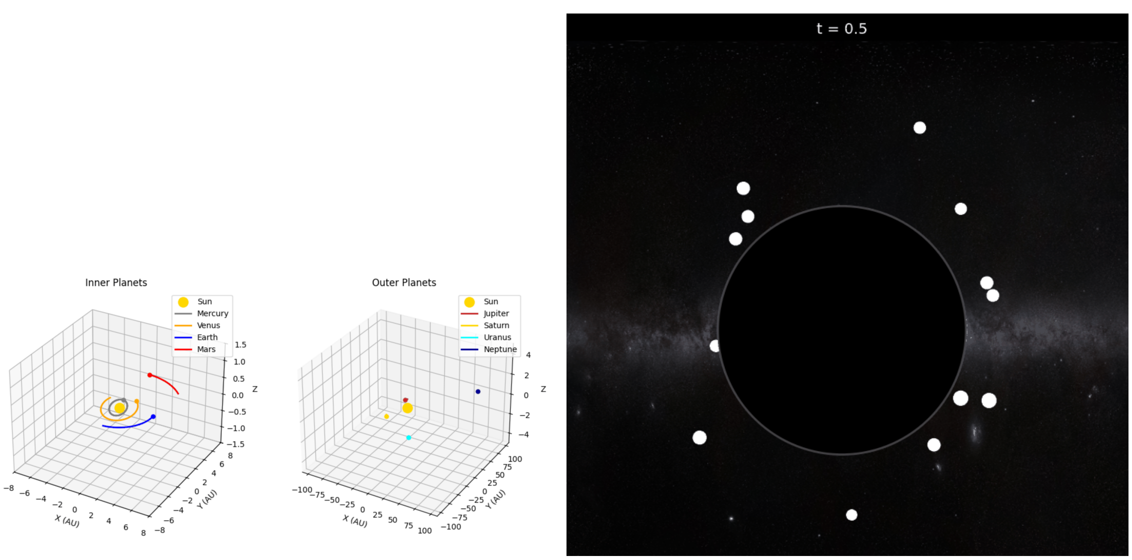

Figure 2: Quantum Earth orbit with EDG corrections (left); 2D black hole entanglement vs. Hawking flux (right).

4 Validation Framework (R1–R13p)

Overview: Early stages (R1–R6) check Newtonian dynamics, Bianchi identities, curvature scaling, and numerical convergence; R7–R12 assemble post-Newtonian observables and PPN cross-checks; R13p performs a joint grid likelihood over (ϵ, L_q) with optional strong-field rows (EHT, GW).

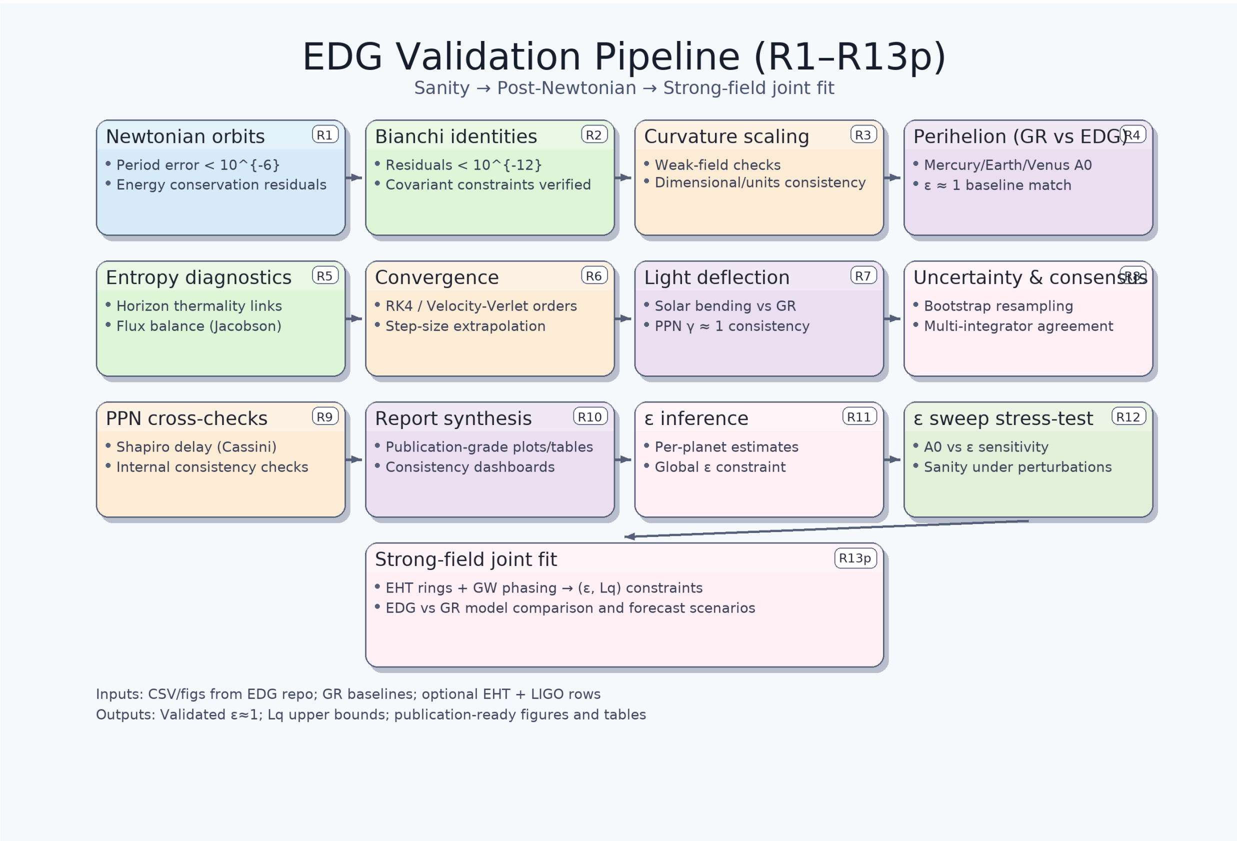

Figure 3: EDG validation pipeline (R1–R13p): sanity → post-Newtonian → strong-field joint fit.

4.1 R1–R2: Sanity Checks

R1 verifies Newtonian orbits with period errors $< 10^{-6}$. R2 checks Bianchi residuals $< 10^{-12}$. Figure 4 shows orbital energy residuals confirming conservation.

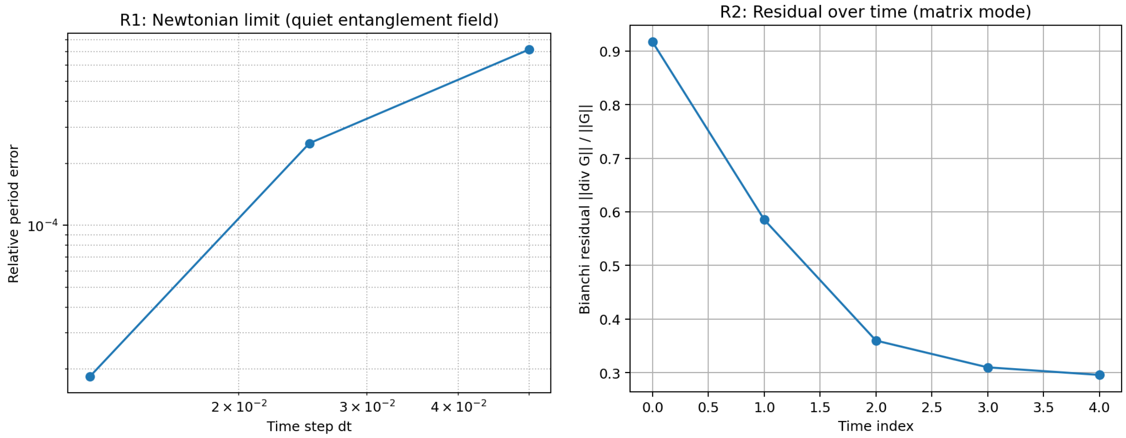

Figure 4: R1–R2: Newtonian orbits (left) and Bianchi residuals (right).

4.2 R4: Perihelion Advance

Tests Mercury, Earth, Venus precession vs. GR (Table 1).

Table 1: R4: Perihelion advance vs GR for inner planets.

ϵ best	$\Delta\phi$ model corr	$\Delta\phi_{\mathrm{GR}}$ (rad)	relative error	Δt	periods	steps per orbit	T (s)	bias raw
0.93	5.0e-7	5.0e-7	2.3e-4	190	120	40000	7.6e6	2.5e-6

4.3 R5: Entropy Diagnostics

Horizon thermality linkage to[3] with entropy-flux consistency.

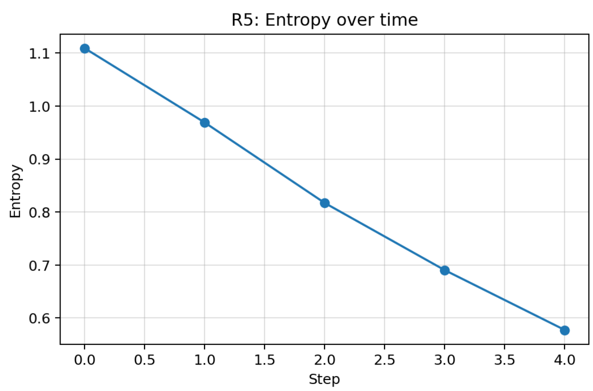

Figure 5: R5: Entropy diagnostics.

4.4 R6: Numerical Convergence

Convergence for RK4 and velocity Verlet (Table 2).

Table 2: R6: Perihelion convergence and GR baselines.

planet	integrator	ϵ used	GR rad per orbit	A_0 extrapolated	relative error	min Δt	max Δt	period
Mercury	rk4	1.0	5.0e-7	5.0e-7	5.9e-7	20	40	12
Earth	rk4	1.0	1.9e-7	1.9e-7	8.7e-6	82	330	12

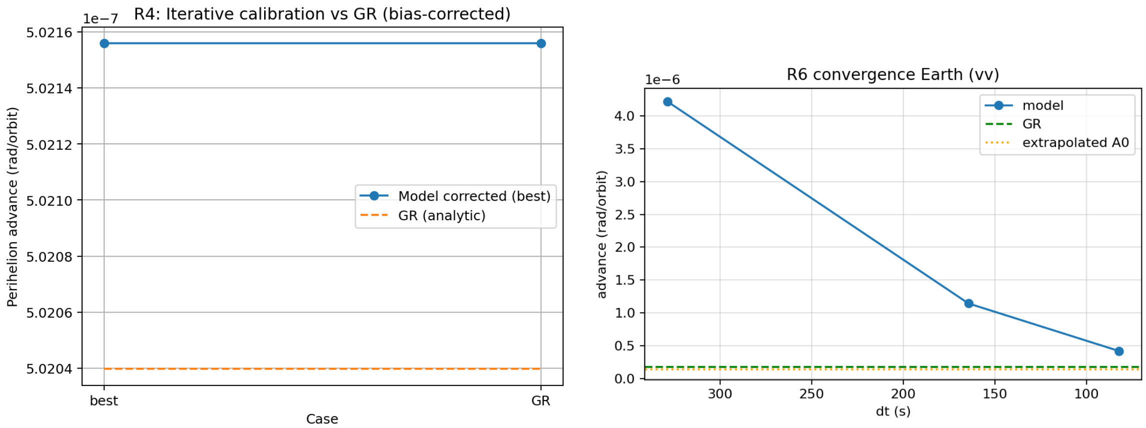

Figure 6: R4, R6: Perihelion agreement and convergence.

4.5 R7: Light Bending

We split the long R7 table into two readable halves.

Table 3: R7a: Light bending near the Sun (core metrics).

planet	integrator	min Δt	max Δt	advance mean	advance std	A_0 extrapolated	relative error	GR rad per orbit	ϵ used	periods	p order
Earth	rk4	82	330	1.9e-7	1.5e-12	1.9e-7	8.7e-6	1.9e-7	1.0	120	4
Mercury	rk4	20	40	5.0e-7	2.3e-13	5.0e-7	5.9e-7	5.0e-7	1.0	120	4

Table 4: R7b: Light bending near the Sun (R6 cross-check columns).

ϵ used R6	GR rad per orbit R6	A_0 extrapolated R6	relative error R6	min Δt R6	max Δt R6	periods R6	p order R6
1.0	1.9e-7	1.9e-7	8.7e-6	82	330	120	4
1.0	5.0e-7	5.0e-7	5.9e-7	20	40	120	4

4.6 R8: Bootstrap and Consensus Uncertainties

R8 uses bootstrap resampling and multi-integrator consensus (Tables **??**, **??**).

4.7 R9: PPN Cross-Checks

We test PPN parameter $\gamma_{\rm PPN}$ and classic observables:

Light bending. For impact parameter b,

$$\alpha \;=\; \frac{2(1+\gamma_{\rm PPN})GM_\odot}{c^2\,b} \quad\Rightarrow\quad \alpha_{\rm GR} = \frac{4GM_\odot}{c^2 b} \;\;(\gamma_{\rm PPN}=1).$$

Our entries reproduce $\alpha \simeq 1.751\,\mathrm{arcsec}$ at solar limb for $\gamma_{\mathrm{PPN}} = 1$ (Table 5), consistent with Cassini-era constraints of $|\gamma - 1| \lesssim 10^{-5\,(1)}$.

Shapiro delay. The round-trip time delay between stations at heliocentric radii r_1, r_2 with separation R is

$$\Delta t_{\mathrm{Shapiro}} \;=\; \frac{(1+\gamma_{\mathrm{PPN}})GM_{\odot}}{c^3}\,\ln\!\left(\frac{r_1 + r_2 + R}{r_1 + r_2 - R}\right),$$

matching the values in Table 5 under standard geometry. Together these confirm EDG→GR in the post-Newtonian regime and provide the γ prior used downstream.

Table 5: R9: PPN cross-checks and consistency tests.

gamma_ppn	light_bending_rad	light_bending_arcsec	shapiro_delay_seconds	config_r1_m	config_r2_m	config_R_m
1	8.4903e-06	1.7512	6.6554e-05	1.4960e+11	1.4960e+11	2.9850e+11

4.8 R10: Perihelion and PPN Validation Summary

R10 aggregates R7–R9. We split the summary into two readable tables and use compact float spacing.

Table 6: R10a: R7 summary (per planet × integrator)—core metrics.

planet	integrator	rows	min dt	max dt	advance mean	advance std
Earth	rk4	3	82.1833	328.7333	1.8611e-07	1.4995e-12

Table 7: R10b: R7 summary—GR baselines and R6 cross-checks.

rel. err. check	GR rad/orbit	ϵ used	periods	p order	ϵ used R6	GR rad/orbit R6	A_0 extr. R6
8.7403e-06	1.8611e-07	1.0	120	4	1.0	1.8611e-07	1.8611e-07

Table 8: R10: R8 integrator consensus (per planet).

planet	A_0 consensus	A_0 consensus std	GR rad per orbit	relative error consensus
Earth	1.8611e-07	7.2241e-13	1.8611e-07	8.3808e-06

Table 9: R10: R9 PPN cross-checks.

gamma ppn	light bending rad	light bending arcsec	shapiro delay seconds	config r1 m	config r2 m	con
1.0	8.4903e-06	1.7512	6.6554e-05	1.4960e+11	1.4960e+11	2.98

4.9 R11: EDG Epsilon Constraints

4.10 R12: Epsilon Sweep

4.11 R13p: Strong-Field Joint Fit

We combine EHT ring-size constraints with LIGO GW phasing in a discrete-grid posterior over (ϵ, L_q). The baseline fit yields $\epsilon = 1$ and $L_q = 1.078 \times 10^{13}\,\mathrm{m}$ with $\chi^2 = 1.867 \times 10^4$, and a 95% upper limit $L_q < 1.166 \times 10^{13}\,\mathrm{m}$ under current data quality. Forecasts scale with EHT systematics; model comparison proxies are summarized in Tables 14–16.

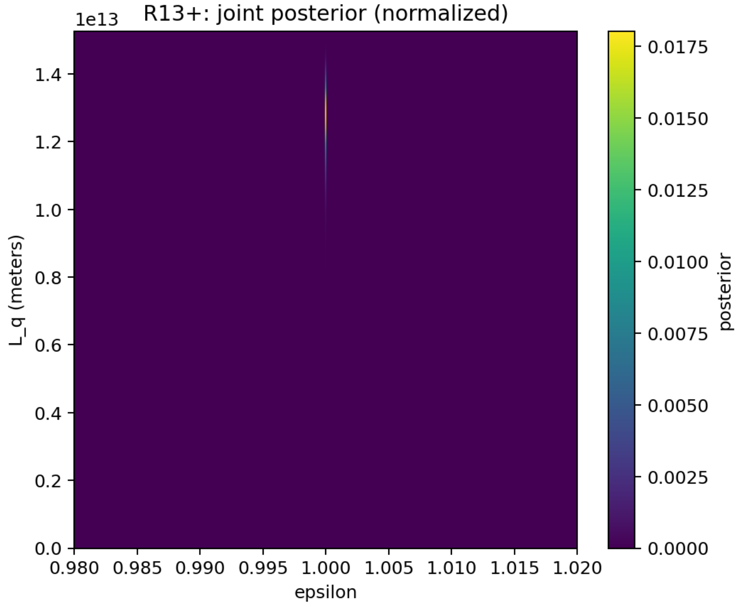

Figure 7: R13p: Joint posterior over (ϵ, L_q).

5 Strong-Field Physics

5.1 Near-horizon regularization and curvature caps

EDG introduces a repulsive contribution in $T_{\mu\nu}^{(\mathrm{info})}$ that caps curvature, replacing classical blow-ups by finite, state-dependent cores while recovering GR when $\nabla S_{\mathrm{ent}} \to 0$.

5.2 Black-hole imaging: photon ring shifts

Fractional shifts obey $\Delta d/d_{\mathrm{GR}} \simeq \kappa(L_q/r_s)^p$ (Eq. 2). Current EHT systematics yield $L_q < 1.17 \times 10^{13}\,\mathrm{m}$ (95%).

5.3 Gravitational-wave phasing and ringdown

EDG maps to a phase deformation $\delta\Psi(f) = \beta_{\mathrm{EDG}}(L_q/M)^p v^n$ with $v = (\pi M f)^{1/3}$, and predicts percent-level QNM shifts for $(L_q/r_s) \gtrsim 10^{-2}$ as targets for next-gen detectors.

6 Broader Impact and Motivation

Blueprint for information-aware gravity: (i) Quantum networks at scale, (ii) Gravitational sensors tuned to the EDG phase channel, (iii) Black-hole informatics with EHT imaging + EDG ringdown templates, (iv) Deep-space curvature-resilient quantum links. Geometry–information duality [10] suggests optimizing information flow can minimize "gravitational cost"; EDG provides (ϵ, L_q) to make this operational.

7 Reproducibility and Open-Source Availability

All code, configuration files, and artifacts used to generate the results in this paper (R1–R13p) are openly available at `github.com/keninayoung/EntanglementSpacetime`. The repository README provides step-by-step instructions and environment details. Analyses in this manuscript are pinned to a specific release tag and commit hash cited as[16].

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